Adding and subtracting fractions

In this sheet you will…

- add fractions with different denominators
- subtract fractions with different denominators
- simplify the answer

Key words

common denominator
LCM · HCF
simplify
numerator · denominator

Method: adding and subtracting fractions

1. **Find the LCM** of the two denominators. This is the common denominator.
2. **Rewrite** each fraction with that denominator: multiply the top **and** the bottom by the same number.
3. **Add or subtract the numerators.** The denominator stays the same: $\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$, not $\frac{5}{16}$.
4. **Simplify** the answer using the HCF, if you can.

Worked example 1 · Add $\frac{3}{4} + \frac{1}{6}$

Step 1

$4 = 2 \times 2$, $6 = 2 \times 3$

LCM $= 2 \times 2 \times 3 =$ **12**

Neither 4 nor 6 is a multiple of the other, so find the LCM. The shared 2 is used once.

Step 2

$$\frac{3}{4} \overset{\times 3}{=} \frac{9}{12} \qquad \frac{1}{6} \overset{\times 2}{=} \frac{2}{12}$$

Rewrite both fractions with denominator 12.

Step 3

$$\frac{9}{12} + \frac{2}{12} = \frac{11}{12}$$

Add the numerators: 9 + 2 = 11. The denominator stays 12.

Step 4

11 is prime and $12 = 2 \times 2 \times 3$

They share nothing, so $\frac{11}{12}$ is already in simplest form.

Answer: $\frac{3}{4} + \frac{1}{6} = \frac{11}{12}$

Worked example 2 · Subtract $\frac{5}{6} - \frac{1}{10}$

Step 1

$6 = 2 \times 3$, $10 = 2 \times 5$

LCM $= 2 \times 3 \times 5 =$ **30**

Neither 6 nor 10 is a multiple of the other. The shared 2 is used once.

Step 2

$$\frac{5}{6} \overset{\times 5}{=} \frac{25}{30} \qquad \frac{1}{10} \overset{\times 3}{=} \frac{3}{30}$$

Rewrite both fractions with denominator 30.

Step 3

$$\frac{25}{30} - \frac{3}{30} = \frac{22}{30}$$

Subtract the numerators: 25 − 3 = 22.

Step 4

$22 = 2 \times 11$, $30 = 2 \times 3 \times 5$, HCF = 2

$$\frac{22}{30} \overset{\div 2}{=} \frac{11}{15}$$

Simplify: divide top and bottom by the HCF.

Answer: $\frac{5}{6} - \frac{1}{10} = \frac{11}{15}$

Exercise write all your steps in your notebook, then the answer here

A **Find the LCM** of the two denominators. Show the prime factors.

1 6 and 9 LCM = _____

2 8 and 12 LCM = _____

3 10 and 15 LCM = _____

4 4 and 7 LCM = _____

B **Add.** Simplify if you can.

1 $\frac{1}{2} + \frac{1}{3} =$ _____

2 $\frac{2}{5} + \frac{1}{4} =$ _____

3 $\frac{3}{8} + \frac{1}{6} =$ _____

4 $\frac{1}{6} + \frac{3}{10} =$ _____

C **Subtract.** Simplify if you can.

1 $\frac{1}{2} - \frac{1}{3} =$ _____

2 $\frac{3}{4} - \frac{2}{5} =$ _____

3 $\frac{7}{8} - \frac{1}{6} =$ _____

4 $\frac{3}{10} - \frac{1}{6} =$ _____

D Sam says: "$\frac{2}{3} + \frac{1}{4} = \frac{3}{7}$, *because 2 + 1 = 3 and 3 + 4 = 7.*" Explain his mistake. Then find the correct answer.

E **Problems.**

1 Leo eats $\frac{1}{4}$ of a pizza and Ines eats $\frac{1}{6}$ of it. What fraction of the pizza is eaten? What fraction is left?

2 Priya spends $\frac{2}{5}$ of her money on a book and $\frac{1}{4}$ on lunch. What fraction of her money is left?

F **Mixed practice.** Simplify if you can.

1 $\frac{2}{3} + \frac{1}{5} =$ _____

2 $\frac{5}{8} - \frac{1}{3} =$ _____

3 $\frac{7}{10} - \frac{1}{4} =$ _____

G **Bigger numbers.** Find the LCM first, showing the prime factors. Simplify if you can.

1 $\frac{5}{12} + \frac{7}{30} =$ _____

2 $\frac{11}{15} - \frac{7}{20} =$ _____

CHALLENGE

Find two different fractions with numerator 1 that add up to $\frac{1}{2}$.

Summary checklist

- ☐ I can find the LCM of two denominators.
- ☐ I can add and subtract fractions with different denominators.
- ☐ I can simplify my answer using the HCF.